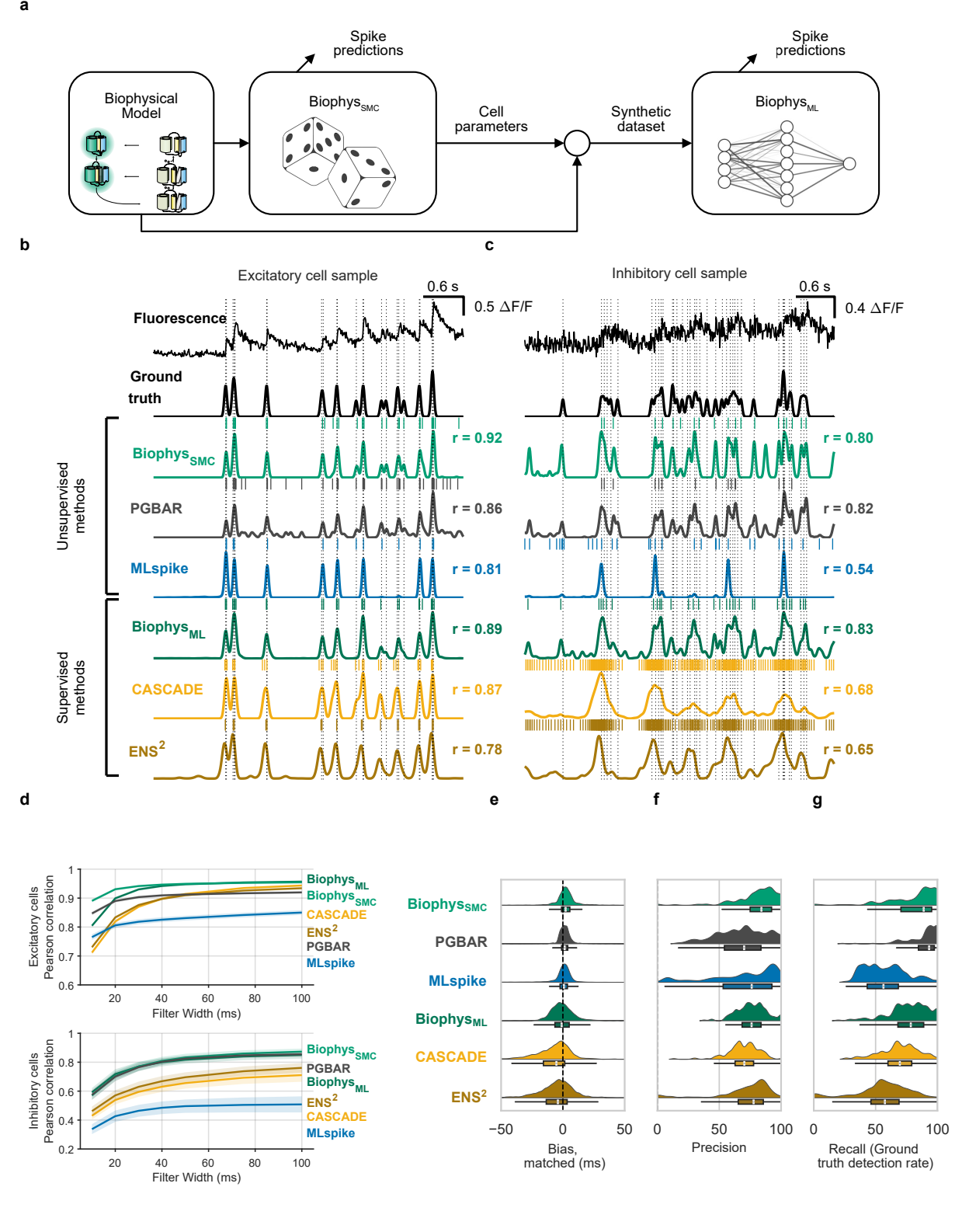

a
Biophysical Model
Spike predictions
Biophys$_{SMC}$
Cell parameters
Synthetic dataset
Spike predictions
Biophys$_{ML}$
b
c
Excitatory cell sample
0.6 s
0.5 ΔF/F
Inhibitory cell sample
0.6 s
0.4 ΔF/F
Fluorescence
Ground truth
Unsupervised methods
Supervised methods
Biophys$_{SMC}$
r = 0.92
r = 0.80
PGBAR
r = 0.86
r = 0.82
MLspike
r = 0.81
r = 0.54
Biophys$_{ML}$
r = 0.89
r = 0.83
CASCADE
r = 0.87
r = 0.68
ENS^2
r = 0.78
r = 0.65
d
e
f
g
Excitatory cells Pearson correlation
Filter Width (ms)
Biophys$_{ML}$
Biophys$_{SMC}$
CASCADE
ENS^2
PGBAR
MLspike
Inhibitory cells Pearson correlation
Filter Width (ms)
Biophys$_{SMC}$
PGBAR
Biophys$_{ML}$
ENS^2
CASCADE
MLspike
Bias, matched (ms)
Precision
Recall (Ground truth detection rate)